MIL-D-9914A(USAF)
6 April 1965

Superseding
MIL-D-9914(USAF)

MILITARY SPECIFICATION

DIAMONDS, INDUSTRIAL

1. SCOPE

1.1 Scope. This specification defines the requirements for mounted and unmounted industrial diamonds used in dressing and truing abrasive wheels.

1.2 *Classification.* Industrial diamonds shall be of the following types and grades (see 6.2).

Type I - Mounted (*see Figure 1*)
Type II - Unmounted

Grade A - Select
Grade B - 1st Quality
Grade C - 2nd Quality
Grade D - 3rd Quality

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MIL-W-17929	Wheels, Abrasive, Precision

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 5350
FSC 3460

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:12Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The stones shall be natural African diamonds. The stones shall not be opaque.

3.2 Grades.

3.2.1 Grade A - Select. The stones shall be clear with no flaws, no naats or other deformities. They shall be octahedral in shape with six good points.

3.2.2 Grade B - 1st Quality. The stones shall be clear with no flaws, no naats or other deformities. They shall be octahedral in shape with at least four good points.

3.2.3 Grade C - 2nd Quality. The stones shall be clear with no flaws, clear naats permissible. They shall be octahedral in shape with at least three good points.

3.2.4 Grade D - 3rd Quality. The stones shall be clear with no flaws, no naats or other deformities. They shall be any shape with at least one good point.

3.3 Mounting. Type I diamonds shall be mounted in one of the standard commercial holders shown in Figure 1, as specified (see 6.2) and shall be set by the sintered powdered metal process. They shall be mounted concentric within 0.002 inches of the centerline of the holder.

3.4 Carat size of diamonds.

3.4.1 Type I diamonds. The average carat size of the diamonds in a lot shall be as specified in Table I. No individual diamond shall vary more than 10% from the specified weight.

3.4.2 Type II diamonds. The average carat size of the diamonds in a lot shall be as specified (see 6.2). No individual diamond shall vary more than 10% from the specified weight.

3.5 Identification marking (Type I). The manufacturer's name and the weight (in carats) of the diamond shall be stamped on the holder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:12Z
Check the source to verify that this is the current version before use.

TABLE I. Carat Size of Type I Diamonds.

TYPE HOLDER (From Figure 1)	GRADE A, B, AND C DIAMONDS (Carat Size)	GRADE D DIAMONDS (Carat Size)
A	0.25	0.10
B	0.25	0.07
C	0.25	0.10
D	0.25	0.10
E	0.25	0.10
F	2.00	0.50
G	1.50,2.00 or 3.00	0.10
H	0.25	0.10
I	0.20 to 0.25	0.07
J	0.25	0.10
K	0.50	0.10
L	0.20	0.10
M	1.00	0.50
N	0.25	0.10

3.6 Capability. When tested as specified in 4.4.1 the diamond shall not lose its point nor shall there be any dimensional changes. The diamonds shall be capable of dressing wheels of the minimum dimensions specified in Table II.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein.

Source: https://assist.dla.mil -- Downloaded: 202640-17T13:12Z
Check the source to verify that this is the currentversion before use.

Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

TABLE II. Diamond Dressing Capability on Abrasive Wheels.

DIAMETER (Inches)	WIDTH OF FACE (Inches)	SIZE OF DIAMONDS (Carats)
1	3/16	0.07
2	1/4	0.10
4	1/2	0.25
7	1/2	0.50
12	1	1.00
14	1	1.50
20	2	2.00
20	6	3.00

4.2 *Sampling for lot inspection.*

4.2.1 Inspection lot. For the purposes of sampling and inspection a lot shall consist of all diamonds of the same type and grade offered for delivery at one time.

4.2.2 Sampling for inspection. A random sample of diamonds shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level 1 and an AQL of 1.0 percent defective.

4.3 Inspection for acceptance. Each of the sample diamonds selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in Table III. Any diamond in the sample containing one or more defects shall be rejected and if the number of defective diamonds in any sample exceeds the acceptance number for that sample the lot represented by the sample shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:12Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of Defects.

Categories	Defects
Critical	None defined.
Major	
101	Opaque diamonds.
102	Diamonds not the specified type.
103	Diamonds not the specified weight.
104	Diamonds contain flaws, naats or other deformities.
105	Diamonds not specified shape.
106	Not specified number of points.
107	Use of unauthorized material.
108	Holder dimensions not in conformance with Figure 1.
109	Identification marking missing.

4.4 Acceptance test.

4.4.1 Capability. One abrasive wheel conforming to Military Specification MIL-W-17929, Type I shall be dressed using standard shop procedures.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing. Unless otherwise specified the diamonds and diamond holders shall be preserved and packed in accordance with good commercial practice.

5.2 Marking. In addition to any marking required by the contract or order the holders shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. Diamonds covered by this specification are to be used in the dressing and truing of abrasive wheels.

6.2 Ordering data. Procurement documents should specify:

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:12Z
Check the source to verify that this is the current version before use.

a. Title, number and date of this specification.

b. Type of diamond required (see 1.2).

c. Grade of diamond required (see 1.2 and 3).

d. Carat size of diamond required (see 3.1).

e. Quantity of diamonds required.

f. If Type I diamond is required the holder needed should be specified from Figure 1.

g. If a holder other than those shown in Figure 1 is required, procurement documents should include a drawing of the desired holder.

h. Preservation and packing requirements, if different (see 5.1).

6.3 Superseding data. The grades of diamonds specified in this document supersede the grades of MIL-D-9914(USAF) dated 19 Jun 1961 as shown in table IV.

TABLE IV Superseding data

MIL-D-9914A(USAF)	Supersedes MIL-D-9914(USAF)
Grade A - Select	Grade WA - Select
Grade B - 1st Quality	Grade WA1 - 1st Quality
Grade C - 2nd Quality	Grade WA2 - 2nd Quality
Grade D - 3rd Quality	Grade WA3 - 3rd Quality

Custodian:

Air Force - 69

Preparing Activity:

Air Force - 69

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:12Z
Check the source to verify that this is the current version before use.